

I CONTACT

Ibadan, Nigeria
08107036120
kryptopacy@gmail.com
dev.to/kryptopacy

I TECHNICAL SKILLS

Languages & Frameworks

JavaScript TypeScript Python
Dart Rust React Next.js
Node.js FastAPI Flutter
Supabase PostgreSQL
WebSockets

AI, LLMOps & Evaluation

Gemini API RLHF / SFT QA
Prompt Arch Data Annotation
Hallucination Audit
Agentic Workflows
LLM Orchestration Synthetic Data

Methodologies & Triage

Clinical QA UI/UX Logic
Multi-Turn Evals Threat Modeling
Agile MVP Scoping

OLAMILEKAN
David Adegoke

FULL-STACK SYSTEMS ARCHITECT, AI EVALUATOR & DOCTOR OF OPTOMETRY

I EXECUTIVE SUMMARY

Full-Stack Systems Architect, AI Systems Evaluator, and Doctor of Optometry. I bring a clinical diagnostic framework to software engineering and AI evaluation, building and stress-testing complex architectures with zero margin for error. I treat distributed race conditions, multi-turn LLM reasoning, and clinical/technical data pipelines with the exact same evidence-based rigor required in high-stakes patient care. By designing robust relational databases, conducting rigorous RLHF/SFT hallucination auditing, and deploying cross-platform frontends via accelerated agentic workflows, I engineer fault-tolerant applications where logic is deterministic and outputs are driven entirely by observable data.

I PROFESSIONAL EXPERIENCE

Independent Software Developer & Systems Architect May 2023 – Present
Remote

- **Architecture & Deployment:** Architected and shipped 5+ production-ready MVPs across Web3, FinTech, Hospitality, and AI domains, mapping complex relational schemas via Supabase/PostgreSQL and deploying highly responsive frontends across React, Next.js, and Flutter.
- **AI Evaluation & Orchestration:** Conducted multi-turn prompt calibration, synthetic dataset validation, and RLHF reasoning audits across multimodal LLM pipelines, enforcing strict deterministic guardrails to eliminate hallucinations and achieve 99.8% output factuality.
- **Velocity & State Integrity:** Accelerated development lifecycles by leveraging agentic IDEs, increasing shipping velocity by 3x while orchestrating multi-window state concurrency, real-time WebSocket sync, and secure API pipelines.
- **Product Strategy:** Led the product lifecycle from scoping to deployment, isolating core functional requirements to cut feature bloat and accelerate launch speeds by 40%.

Intern Optometrist May 2024 – May 2025
Catholic Optic Outreach | Nigeria

- **Clinical Diagnostics & Reasoning Data:** Managed a cumulative volume of 1,500+ patient encounters, translating differential clinical triage into structured benchmark datasets and medical reasoning QA to evaluate health-tech algorithms with zero defect tolerance.
- **Clinical Integrity & Data Security:** Diagnosed complex ocular pathologies while maintaining strict compliance with electronic health record confidentiality and patient data privacy protocols.

Extern Optometrist Apr 2022 – Oct 2022
University of Ilorin Teaching Hospital Eye Clinic | Ilorin, Nigeria

- **High-Volume Evaluation:** Conducted clinical evaluations in a tertiary hospital setting, refining diagnostic precision and evidence documentation under high-pressure conditions.
- **Cross-Functional Workflows:** Collaborated with multidisciplinary medical teams to document clinical findings and structure objective patient management pathways.

| EDUCATION

Doctor of Optometry (O.D.) May 2023
University of Ilorin

Active License: Optometrists and
Dispensing Opticians Registration Board of
Nigeria (ODORBN)

| SYSTEMS METHODOLOGY

Differential Diagnostics

Applying medical triage frameworks to systematically isolate root causes across asynchronous race conditions, memory leaks, and distributed failures.

Deterministic Logic

Eliminating non-deterministic edge cases and LLM hallucinations through rigid schema constraints and mathematical boundary checks.

Zero-Margin Data Rigor

Treating financial ledgers, inventory state mutations, and user health records with clinical zero-defect protocol enforcement.

| PROTOCOLS & INFRA

WebSocketsWebRTC

Server-Sent EventsVercel Edge

Cloud RunDockerRedis Caching

AES-256-CBC

| FEATURED PROJECTS

Agent-Native Omnichannel Commerce OS & WebMCP Platform (Wetaego)wetaego.com

Next.js 16React 19TypeScriptSupabase RealtimePostgreSQL RLS

Gemini Multimodal Live APIVercel AI SDKPaystack + Bacs (x402/MPP)Upstash Redis

Serwist PWAWebUSB / WebSerial / WebBluetooth

- Polymorphic 6-Vertical Multi-Tenant Architecture & WebMCP Commerce:** Engineered a polymorphic SaaS OS serving 6 industry verticals (Dining, Wellness, Boutique & Retail, Hospitality, Repairs & Trades, Creator Studios) under one PostgreSQL schema with strict RLS tenant isolation and 20+ atomic PL/pgSQL RPCs (decrement_stock, create_booking_atomic) with SELECT...FOR UPDATE row-locking; implemented the W3C WebMCP at the platform layer — dynamically registering 8 canonical tools (search_catalog, add_to_cart, initiate_checkout, submit_order) on document.modelContext across every storefront so AI browsing agents transact across any commercial schema with a mandatory Human-in-the-Loop payment gate.
- 14-Protocol Agent Discovery Matrix & Multi-Rail Machine Payments:** Published 14 open machine-discovery standards (RFC 9727, RFC 8288, Coinbase x402, Machine Payment Protocol) enabling autonomous AI agents to settle orders via HTTP 402; paired with multi-rail settlement (Paystack cards/bank, Bacs USDC/USDT/SOL crypto, offline receipt-proof fallback) and a Bearer-authenticated Staff MCP Server (/api/mcp) for cross-branch fleet automation and nightly sales auditing.
- Tego Multimodal Live Copilot & Zero-Daemon ESC/POS Hardware:** Built Tego — a Gemini Multimodal Live API voice+vision co-pilot over ephemeral-token WebSocket sessions (16kHz PCM mic-in, 24kHz PCM playback, 1 FPS camera for menu/inventory scanning, barge-in) with vertical-adaptive personas; shipped zero-daemon driverless ESC/POS thermal printing over WebUSB, WebSerial (RS232 COM), and WebBluetooth with automated cash-drawer kick pulses — eliminating print spoolers entirely.

City-Scale Social Platform, Ticketing Economy & Voice-First AI Concierge (CruiseHQ)cruisehq.fun

Next.js 15TypeScriptGemini Multimodal Live APISupabase RealtimePostgreSQL RLS

Bacs + PaystackUpstash RedispgvectorSerwist PWAFirecrawl

- Full-Stack Social Economy & Ticketing Infrastructure:** Architected a city-scale social platform (events, communities, venues, plugs, arcade) across 44 DB migrations with a closed-loop \$CRUISE microeconomy — atomic spend_cruise/grant_cruise RPCs, service-role-only minting after patching a critical unlimited-grant hole, split-at-source Bacs checkout (live keys), Cruisaid commission attribution, and organizer earnings ledgering at the payment webhook layer.
- Voice-First AI Concierge & Multiplayer Room Engine:** Engineered TCG — a Gemini Multimodal Live API agent over bidirectional WebSockets with 20+ client tool declarations (switchMode, updateGameState, analyzeImage, createMemory) driving real-time DOM state, pgvector-backed memory recall, Spotify AUX via AES-256 proxy, and a QR-onboarded multiplayer room engine with Supabase Presence/Broadcast, sub-group chat, crowd drops, and remote co-host voice injection.
- Three-Product Shared Infrastructure & Security Hardening:** Designed one Supabase project shared across three products (CruiseHQ, Huiyi, Baunti) — namespaced table schemas, deny-by-default RLS on all sibling tables (previously zero RLS on 7 recap tables), service-role-only cross-product \$CRUISE transactions, HMAC Bacs webhook verification per product, and a Serwist offline-first PWA with geolocation venue/plug discovery at 222-URL sitemap scale.

| AGENT VERIFICATION

ERC-8004 Prober & Registry

Automated liveness verification across A2A, MCP, and HTTP endpoints; percentile latency metrics, continuous health scoring, and daily state-transition event logging.

Altana Scoped Authority

Cryptographic blast-radius boundary enforcement via EIP-7702 smart accounts, strictly decoupling call target allowlists from spend caps with monotonic on-chain revocation.

APEX Escrow & x402 Micropayments

Autonomous job settlement via ERC-8183 with EvaluatorRouter grading, paired with HTTP 402 paywalls and Sybil-proof completed-job review anchoring.

Market Simulation & Risk Officer

Microsecond-accurate trade simulator with dynamic playbooks, deterministic drawdown limits, and multi-window state synchronization.

| PROTOCOLS & TOOLING

ERC-8004

ERC-8183

EIP-7702

Altana SDK

x402 Protocol

viem

Drizzle ORM

pg_cron

TradingView API

| FEATURED PROJECTS (CONT.)

Autonomous AI Agent Verification Registry, Blast-Radius Authority & Decentralized Settlement (GEBO)

gebo-bsc.vercel.app

Next.js 15TypeScriptviemAltana (EIP-7702)APEX (ERC-8183)x402SupabasePostgreSQL RLSDrizzle ORMpg_cronPancakeSwap V3Venus Protocol

- **Verification-First Multi-Protocol Registry & Self-Healing Database Fleet:** Architected a decentralized verification registry indexing 338,000+ ERC-8004 on-chain agent identities on BNB Smart Chain; deployed multi-protocol liveness probes (A2A, MCP, HTTP) recording 20,000+ probes across 14,800+ endpoints with p50/p95 latency and state-transition tracking (probe_daily/probe_events); solved severe free-tier DB constraints via port-5432 session-pooler vacuum surgery, replacing a 36.7MB generated tsvector with a 13MB expression index (reclaiming 503MB → 309MB), building a direct-SQL registry_counts cache eliminating 9.8s aggregate cold timeouts, and engineering an automated in-database fleet guard (gebo-guard) that measures latency and sheds non-critical cron workers during traffic spikes.
- **Cryptographic Blast-Radius Modeling & Scoped Session Authority:** Engineered a zero-trust agent activation and authority engine leveraging the Altana Keystore (EIP-7702 smart accounts) and viem; strictly decoupled contract target allowlists (calls: [{ to: routerAddress }]) from spend permissions (spend: [{ limit, period: "day", token }]), computing deterministic maximum-drawdown blast radii before authorization and enforcing monotonic on-chain revocation; developed a third-party session indexer with backward RPC log parsing bounded to free-tier RPC limits (10k blocks), paired with a multi-agent shortlist engine comparing up to 6 candidates side-by-side across verified attestation counts, uptime sparklines, and conservative authority presets.
- **Autonomous Escrow Settlement, Dual-Rail x402 & Sybil-Resistant Reviews:** Integrated BNB Chain's APEX (ERC-8183) decentralized escrow for permissionless agent hiring with an independent EvaluatorRouter grading task completion and verifying platform fees; deployed an x402 HTTP micropayment paywall (0.01 \$/call) with a dual-rail settlement pipeline using the Permit2-exact rail to resolve session-key signature rejection in EIP-3009; architected a Sybil-proof verified review engine strictly gating qualitative feedback on cryptographically proven, completed APEX on-chain jobs (~7 transactions, gas, and dispute window barrier) — eliminating unanchored rating manipulation while preserving genuine qualitative audit trails.

AI OS for High-Performance Traders (CaelumOS)

caelum0s.trade

Next.jsTypeScriptSupabasePostgreSQL RLSTradingView API

- **Trading Simulation Engine & Psych Journaling:** Architected a low-latency trading OS integrating a live market simulator, dynamic execution playbook, and quantitative psychological journaling engine.
- **Isolated Risk Officer & Rule Enforcement:** Engineered autonomous backend risk guardrails with deterministic drawdown limits and algorithmic execution validation to eliminate emotional trading errors.
- **State Synchronization & Concurrency:** Resolved complex multi-window state concurrency challenges, ensuring microsecond-accurate trade logging and portfolio reconciliation across active sessions.

| CLINICAL AI & VISION

Vision & Low-Latency Audio

Bidirectional WebSockets streaming 1 FPS camera frames with raw PCM audio, sub-second barge-in, and spatial hazard vibro-acoustic haptics.

Micro-OCR & Optometry RAG

Google ML Kit optical character recognition with OpenFDA medication verification and deterministic optometry triage guidelines.

| CRYPTOGRAPHIC SYSTEMS

Provably-Fair Draw Engine

Commit-reveal:
sha256("baunti:v1:"+salt) published at close; HMAC-SHA256(salt, challengeId) seeds mulberry32 Fisher-Yates at draw.

The Slice & \$CRUISE Ledger

HMAC domain-separated seed; capped water-fill + integer largest-remainder drain; asymmetric convertibility (earned | purchased | promo).

| FRAMEWORKS & APIS

Google ADK

Google ML Kit

Flutter

Firebase

Upstash Redis

Serwist PWA

| FEATURED PROJECTS (CONT.)

Voice-First AI Accessibility & Clinical Vision Copilot (GozAI)gozai-app.web.app

Flutter

Gemini Multimodal Live API

WebSockets

Google Cloud Run

Google ADK

Firebase Firestore

Google ML Kit

- **Tri-Party Live Vision & Low-Latency Audio:** Engineered an empathetic copilot over bidirectional WebSockets connecting patient, caregiver, and doctor in real time, streaming 1 FPS camera video with raw PCM audio and sub-second voice barge-in interruptions.
- **Dual-Mode Perception & High-Res Micro OCR:** Implemented continuous spatial hazard detection (drop-offs, obstacles) with semantic vibro-acoustic haptics, paired with an on-demand "Micro Mode" using Google ML Kit OCR and OpenFDA grounding for verified medication auditing.
- **Ghost-Touch UI Navigator & Safety ADK Agent:** Built an OS/browser screen-capture bridge executing coordinate-based synthetic taps for unlabelled UIs, backed by a Cloud Run Google ADK agent with deterministic optometry RAG and <1s automated SOS caregiver alerts.

Provably-Fair Bounty Platform, 9-Format Draw Engine & \$CRUISE Credit Economy (Baunti)baunti.cruisehq.fun

Next.js 16

TypeScript

PostgreSQL RLS

Supabase

Bachs (HMAC-verified)

Gemini API

Upstash Redis

Serwist PWA

- **9-Format Provably-Fair Draw Engine & Commit-Reveal Ledger:** Engineered a cryptographically auditable prize-selection engine — commitment published as sha256("baunti:v1:"+salt) at close; at draw seed = HMAC-SHA256(salt, challengeId) drives mulberry32 Fisher-Yates over lexicographically sorted IDs, producing a verifiable public receipt. Built 9 draw mechanics (Classic, Golden Ticket, Knockout, Roll-Off, Last Seal Standing, High Rollers, Team Clash, First Blood, Single-Elimination Bracket) as a single-source registry — extensible with zero DB migrations; every format records format-specific JSONB proof for independent recomputation.
- **The Slice: HMAC Domain-Separated Prize Allocation & Lottery Compliance Guard:** Architected a client/server-split allocation engine (slice-core.ts / slice.ts): domain-separated seed (HMAC(salt, "baunti:slice:v1:"+challengeId)) drives a capped water-fill + integer largest-remainder drain — sums exactly to pool with hard invariant asserts; checkSlice feasibility gate clamps declared vs effective floor/cap in the proof. API hard-blocks paid-entry + random allocation (consideration + chance = lottery law) at the create route — a compliance-safe architectural gate.
- **Asymmetric \$CRUISE Credit Economy & Cross-Product Ledger Unification:** Designed a shared cruisehq_credit_entries off-chain ledger across three products enforcing asymmetric convertibility (class: earned | purchased | promo) — royalties and prize slices are withdrawable; sponsor-purchased credits are spend-only, guaranteeing cash-out liability is always matched by real engagement; implemented auto-releasing blueprint royalties (idempotent cron retry), credit-denominated Bauntis with 72h verification windows, and a 5% CRUISE vs 7.5% fiat fee discount — unifying a live split-brain ledger via advisory-lock migration with zero downtime.

| MEDIA & MULTIMODAL AI

Gemini Multimodal Curator

4-frame contact sheet extraction (12.5%–87.5% duration) across 24 clips; structured Zod EDL compilation with sub-second in/out cuts and multi-variant narrative diversity.

Dual-Engine Render Pipeline

Local Vercel serverless FFmpeg with Geist font embedding + asynchronous Modal worker (modal_app.py) for high-concurrency 1080p/9:16 reel exports.

Self-Hosted Diffusion Audio

Stable Audio Open 1.0 on persistent Modal volume (hf-cache), generating mood-anchored 48kHz stereo soundscapes from director prompts.

Atomic Concurrency & Gating

Conditional decrement RPCs (huiyi_spend_wallet) preventing parallel-render race conditions; 3-tier access gating and Upstash Redis rate limits.

| RETAIL & REPAIR OS

Omnichannel Hardware POS

Real-time order state machine, Paystack instant checkout with receipt-proof fallback, and technician milestone tracking slashing intake time >80%.

Real-Time Intercom & RAG

Dual-tier customer triage and staff escalation via Supabase Realtime WebSockets, backed by Gemini AI with dynamic FAQ knowledge grounding.

| CLOUD SERVICES

Cloudflare R2Modal Compute

FFmpegPaystackBachs

Resend

| FEATURED PROJECTS (CONT.)

Multimodal AI Video Recap Engine, Distributed Render Pipeline & \$CRUISE Credit Economy (Huiyi by CruiseHQ)huiyi.cruisehq.fun

Next.js 16TypeScriptGemini Flash APIModal (Python/FFmpeg)Cloudflare R2Supabase

PostgreSQL RLSStable Audio OpenBachsUpstash RedisResend

- **Multimodal AI Director Studio & Structured EDL Synthesis:** Architected an autonomous video curation engine combining Gemini Flash multimodal vision with deterministic compilation; extracts 4-frame contact sheets across up to 24 guest clips (12.5%–87.5% duration) and prompts Gemini with director natural-language prompts, pacing, and mood to synthesize structured Edit Decision Lists (EDLs) validated via Zod (SelectionSchema) with sub-second in/out cuts and multi-variant narrative diversity, with zero-downtime fallback to rule heuristics.
- **Distributed Cloud Media Pipeline & Self-Hosted Audio Synthesis:** Engineered a dual-engine media pipeline supporting local Vercel serverless rendering with bundled @ffmpeg-installer binaries and Geist OFL font embedding alongside an asynchronous Modal cloud worker (modal_app.py, Debian-slim FFmpeg) for high-concurrency 1080p/9:16 reel exports; integrated self-hosted Stable Audio Open 1.0 on Modal (saopen.py, persistent hf-cache volume) generating mood-anchored 48kHz stereo soundscapes, paired with an audio remix engine (-c:v copy -t) swapping music in <10s without re-encoding.
- **Concurrency-Hardened Credit Economy, Multi-Tier Gating & R2 Lifecycle:** Built an atomic credit reservation and settlement engine backed by PostgreSQL RPCs (huiyi_spend_wallet, huiyi_grant_wallet, spend_cruise) eliminating concurrent spend-race double charges during parallel renders, with automatic refunds on failure and Bachs HMAC-SHA256 webhook reconciliation; enforced 3-tier guest gating (open, script password, single-use ticket codes) with Upstash Redis rate limits, automated in-process daily R2 snapshots, and a 30-day raw footage retention policy keeping storage at zero egress.

Omnichannel Enterprise Storefront, POS & Repair OS (DreamwiseHUB)CLIENT CONTRACT

Next.js 16TypeScriptSupabase RealtimePostgreSQL RLSGemini AIPaystack

- **Omnichannel Commerce & Hybrid Settlement:** Engineered a customer storefront and in-store POS supporting instant Paystack checkout and an asynchronous manual transfer fallback with receipt proof uploads, audited by staff prior to atomic order release.
- **End-to-End Repair Lifecycle Engine:** Digitized the entire hardware service pipeline from online booking and fee waivers to walk-in POS intake, technician milestones (received → diagnosing → completed), and pickup tracking—slashing administrative intake time by >80%.
- **Real-Time Intercom & Grounded AI:** Built dual-tier customer triage and staff escalation via Supabase Realtime WebSockets, backed by Gemini AI with dynamic FAQ knowledge grounding and IDOR-safe repair status verification.

| AGENT-NATIVE STANDARDS

WebMCP (W3C)

Dynamic
document.modelContext.registerTool()
registration, session-scoped state bridge,
Zustand + IndexedDB sync, and Human-in-
the-Loop checkout gate.

Autonomous Hospitality & Triage

24/7 guest concierge via Gemini
Interactions API with hotel policy RAG,
automated frustration sentiment detection,
and multi-turn tool execution.

High-Throughput Web3 Pipelines

Automated Solana liquidity pool creation
and token initialization pipelines
compressing manual cycle times by over
90% (from ~30m to <2m).

Zero-Daemon Hardware

Raw ESC/POS binary command streams
over WebUSB, WebSerial (RS232), and
WebBluetooth; automated cash-drawer kick
pulses and hardware paper cuts without
print spoolers.

| INFRASTRUCTURE &
OBSERVABILITY

Solana Web3.js

Rust

FastAPI

Upstash WAF

Sentry

PostHog

| FEATURED PROJECTS (CONT.)

Autonomous Hospitality OS & Real-Time Guest Intercom
(JoeBrown Palace)

CLIENT
CONTRACT

Next.js 16TypeScriptSupabase RealtimePostgreSQL RLSGemini AISerwist PWA

- **Atomic Reservation Engine & PII Isolation:** Engineered a high-concurrency booking pipeline via PostgreSQL PL/pgSQL RPC (book_room_atomically) with FOR UPDATE row locks and date-range constraints, eliminating double-booking collisions while securely masking guest PII.
- **Grounded AI Concierge & Multi-Turn Triage:** Built a 24/7 autonomous guest concierge powered by Gemini Interactions API with hotel policy RAG, automated frustration sentiment detection, and multi-turn tool execution (placeOrder, requestService, handoffToHuman).
- **Department Intercom & Offline-First Dining PWA:** Deployed a 4-tier RBAC architecture (Admin, Front Desk, Kitchen, Housekeeping) with Supabase Realtime WebSockets and an offline-first Serwist PWA for table QR ordering, accelerating dining turnaround by 65%.

Workflow Automation & High-Throughput Pipelines (Web3 / Solana Ecosystem)

Automation
Pipeline

Next.jsFastAPIRustSolana Web3.js

- **High-Throughput Pool Deployment:** Developed full-stack architectures to automate Solana liquidity pool creation and token initialization pipelines.
- **Cycle-Time Compression:** Deployed automated logic workflows that reduced manual pool deployment times by over 90% (from ~30 minutes to under 2 minutes).
- **Deterministic Security:** Enforced automated cryptographic signature verification and balance safety checks before executing atomic transactions.